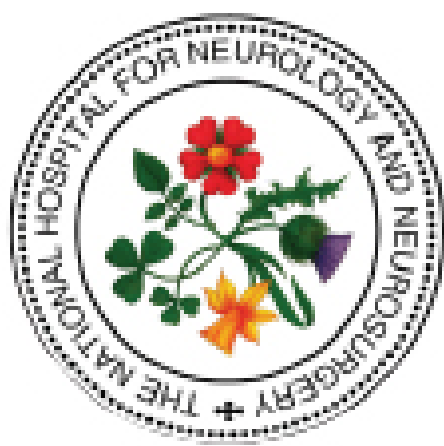




CLONIDINE for SEDATION and CO-ANALGESIA in NCCU

INDICATIONS FOR USE

Clonidine may be used as an unlicensed treatment with consent of the NCC in the following circumstances:

- Alcohol or opiate withdrawal reactions
- Agitation with tachycardia and hypertension, especially during weaning from the ventilator
- Sympathetic 'storming' after brain injury
- As additional sedative when adequate sedation not maintained using standard drugs

Paroxysmal Sympathetic Hyperactivity (PSH) - *also commonly referred to as 'dysautonomia' or 'sympathetic storming'* is characterised by agitation, diaphoresis, hyperthermia, hypertension, tachycardia, tachypnoea accompanied by motor symptoms such as hypertonia and extensor posturing.

Clonidine acts primarily on the central nervous system, resulting in reduced sympathetic outflow and decrease in HR and BP.

Peak action occurs in 10 minutes and lasts for 3-7 hours after single intravenous dose. Oral administration reaches a peak plasma level within 60-90 minutes.

COMPATABILITY

- Compatible with 0.9% sodium chloride and 5% dextrose
- Reduction in BP can be potentiated by concurrent administration of other hypotensive agents

SIDE EFFECTS

BRADYCARDIA	CONFUSION
HYPOTENSION	DEPRESSION
SEDATION	NIGHTMARES
HEADACHE	CONSTIPATION
	DRY MOUTH

HYPERTENSIVE CRISIS may occur if stopped abruptly - *especially if patient has been on large dose or for >1 week*

Agitation can also occur with abrupt withdrawal

Clonidine should be used with extreme caution in patients with hypotension, bradycardia, low cardiac output or impaired ventricular function.

AVOID OVER-STIMULATION

May require individual patient parameters to avoid over-treatment with sedation

Early mobilisation / tilt table / environmental modification may also help to modulate paroxysmal sympathetic hyperactivity

LIGHT SEDATION in MECHANICALLY VENTILATED PATIENTS to FACILITATE WEANING and EXTUBATION

PRESENTATION

1 Ampoule of clonidine hydrochloride contains **150mcg in 1ml** solution for injection

STEP 1

ADMINISTRATION

Dilute **600mcg** of clonidine (4 ampoules of 150mcg/mL) with 0.9% sodium chloride to produce a solution of **600mcg in 30ml** (20mcg/mL)

Deliver by syringe pump either peripherally (*via **dedicated peripheral intravenous line***) or centrally

STEP 2

BOLUS DOSE

An initial **bolus dose of 60mcg** (3mL of 600mcg/30mL solution) may be given by slow intravenous injection **over 10-15 minutes** to avoid possible transient hypertensive effect

STEP 3

INFUSION

Follow with a continuous infusion rate of **0.5 – 2mcg/kg/h**

Titrate infusion rate as BP and HR allow until **symptom control**

Consider reduction in other sedatives and opioids

Higher doses may be needed up to **maximum 4mcg/kg/h**, after d/w NCC consultant

STEP 4

WEANING

When no longer required WEAN to avoid rebound hypertension / agitation

- ☐ Sudden withdrawal may result in agitation, sweating, hypertension
- ☐ Reduce dose gradually – *rate of reduction will depend on duration of infusion*
- ☐ Withdrawal usually over several hours or longer if infusion has lasted >1 week
- ☐ Reduce rate by **0.25 – 0.5mcg/kg/h** to minimise rebound hypertension / agitation (*slower rate of dose reduction may be required for some patients*)

Conversion to oral clonidine from intravenous clonidine

- ☐ Calculate total dose of IV clonidine given over 24 hours
- ☐ Divide total daily dose into 3 doses to be given orally up to **maximum 600mcg/tds**
- ☐ Doses reduced gradually (25mcg increments) according to response

DESIRED DOSE mcg/kg/h	INFUSION RATE (mL/h) of 20mcg/mL SOLUTION						
	PATIENT WEIGHT (kg)						
	40	50	60	70	80	90	100
0.5	1ml/h	1.25ml/h	1.5ml/h	1.75ml/h	2ml/h	2.25ml/h	2.5ml/h
1.0	2ml/h	2.5ml/h	3ml/h	3.5ml/h	4ml/h	4.5ml/h	5ml/h
1.5	3ml/h	3.75ml/h	4.5ml/h	5.25ml/h	6ml/h	6.75ml/h	7.5ml/h
2	4ml/h	5ml/h	6ml/h	7ml/h	8ml/h	9ml/h	10ml/h